Outpost Zero (Remote Arctic / Extreme Environment, Limited Resources)

Executive Summary

Outpost Zero is a harsh **Arctic battlefield** built for **extreme-environment warfare. limited-resource logistics. survival-driven autonomy. and sensor-denied combat**. The terrain features frozen tundra. ice ridges. research domes. buried bunkers. geothermal vents. supply caches. and wind-blasted plateaus. The map emphasizes Obsidian Protocol's autonomous cold-weather logic. resource-rationing behaviors. sensor-fusion under blizzards. and logistics under severe environmental constraints. In AR (Nreal Light/Air). the map anchors at ~4–5 m viewing distance. covering a 10×10 m physical area representing ~1 km² of Arctic terrain. Key features include whiteout zones. limited-resource supply nodes. frozen chokepoints. and survival-driven tactical decision-making.

Design Goals & Gameplay

Extreme-Environment Warfare

Outpost Zero signifies cold-weather hazards: freezing temperatures. ice storms. low visibility. and equipment degradation. Units must adapt to environmental threats as much as enemy fire.

Limited Resources & Survival Logistics

Fuel, ammunition, and power are scarce. Controlling geothermal vents, supply caches, and research domes determines long-term survivability.

Autonomy & Survival Pathfinding

Autonomous units choose routes based on wind exposure, ice stability, thermal cover, and resource availability. Units avoid whiteout zones unless necessary.

Sensor & Information Warfare

Blizzards, ice fog, thermal distortion, and magnetic interference degrade sensors. Relay towers on ridges restore clarity. Destroying relays creates massive blind pockets across the tundra.

Logistics & Resources

Fuel caches. geothermal stations. and research domes serve as resupply nodes. Convoys must navigate frozen terrain and avoid ambush-prone ice corridors.

Deployment & Progress

A fixed deployment budget (~30,000 points) forces players to choose between survival-optimized infantry, cold-weather drones, or heavy Arctic armor. Blue deploys from the southwest landing zone; Red deploys from the northeast geothermal command ridge.

Spatial Constraints (Nreal AR)

Device Capabilities

Nreal Light/Air: ~52° diagonal FOV, 1920×1080 per-eye resolution. Optimal viewing distance ~4–5 m.

Playable Area & Scaling

10×10 m physical area = 1 km² Arctic zone (1:100 scale). Designed for tabletop AR.

Tracking & Occlusion

Inside-out tracking detects horizontal planes. Depth-mesh (if available) provides limited occlusion. Virtual objects render over real-world view; assume minimal real-world occlusion.

Layout Overview

Coordinate System

1000×1000 grid (1 unit = 1 m). Origin (0,0) = southwest corner.

Key Locations

Blue HQ Landing Zone Ground — (0–0) 150×150 m Snow berms. open crates. cold-weather shelter area.
Red HQ (Geothermal Command Ridge) — (1000,1000). 150×150 m Heated command bunker with radar mast and thermal relays.
Central Ice Basin — (500,450). 300×300 m Frozen lake. unstable ice. low visibility. lone-ranger fire lanes.
Research Dome Cluster — (750,800). 170×130 m Lab. generators. thermal chokes. interior cover.
Buried Bunker Network — (800,300). 200×200 m Underground tunnels. firing ports. hardened defensive positions.
Whiteout Corridor — (0,450 → 400,900 Severe blizzard zone: extreme sensor degradation.
Frozen Trail Route — (300,0 → 450,350 Safer but slow: ice cracks and wind exposure.
Thermal Relay A — (650,150). +240 m Lone-ranae sensor vantage point.
Thermal Relay B — (150,650). +360 m Comms relay hub and aeothermal control node.
Neutral Forward Outpost — (500,950) Drone resupply zone.

Chokepoints & Sightlines

Whiteout Corridor

Severe visibility loss; ideal for stealth ambushes and survival-driven tactics.

Buried Bunker Network

Interior flanking routes; extremely strong defensive positions.

Ice Basin

Unstable terrain; long-range engagements with environmental hazards.

Frozen Trail Route

Predictable but safe; vulnerable to overwatch from ridges.

Sightlines

Thermal Relay A provides long-range sensor coverage (~300 m). Blizzards and ice fog create fog-of-war pockets.

Deployment Zones

Blue Deployment: Ski landing zone. cold-weather shelter.
Red Deployment: NE aeothermal command ridge + thermal defenses.
Neutral Forward Zone: Outpost at (500,950) for drone drops.

Assets & Environment

Terrain

Ice ridges, frozen lakes, snow berms, geothermal vents, buried bunkers.

Vegetation

None — Arctic tundra only.

Buildings & Props

Military: Radar dishes, thermal relays, bunkers. Industrial: Generators, pipelines, research domes. Logistics: Trucks, crates, fuel drums, drones.

Effects

Snowstorms, ice fog, wind gusts, thermal distortion.

Audio

Wind, cracking ice, distant machinery, radio chatter.

LOD & Budget

Asset Cate	Count	Tri/Item	Total Tris
Ice Ridges	3	200k	600k
Bunkers/I	4	80k	320k
Buildings/	15	50k	750k
Roads/Pa	4	30k	120k
Props	50	10k	500k
Total	—	—	**~2.29M**

Texture Budget

2048² for terrain/buildings, 1024² for props. DXT1/5 compression. Target ≤100 MB.

Unreal / Nreal Integration

Engine & Plugins

Unreal ARTemplate + Nreal SDK (NREAL). ARKit/ARCore enabled.

Rendering

Forward renderer, stationary lights, baked GI, aggressive culling.

Anchors

Single spatial anchor for entire map.

Interaction

Phone pointer for pan/zoom; HUD as screen-space widgets.

Multiplayer

Client-server architecture; offload AI/physics to server.

Implementation Plan & Testing

Milestones

1. Prototype Layout — Month 1 Block-out ice basin. bunker network. research domes. AR anchor test.
2. Core Systems — Month 2 NavMesh. cold-weather autonomy behaviors. survival logic.
3. AR Interaction — Month 3 Anchors. plane detection. basic interactivity.
4. Content Fill — Month 4–5 Final models. textures. lighting. audio.
5. Gameplay & Polish — Month 6–7 Objectives. balance. UGV integrations. ice-fog. thermal distortion.
6. Testing & QA — Month 8 Multiplayer stress test. sensor validation.

Testing Plan

Autonomy

Verify cold-weather navigation, survival logic, hazard avoidance.

Comms/Sensors

Test relay destruction, fog-of-war updates, LOS checks under blizzard conditions.

Performance

≥60 Hz AR render on target device.

QA Checklist

Navigation, survival logic, deployment budget enforcement, anchor drift, sensor accuracy.